FOUNDATION COURSE

START YOUR
PREPARATION
FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 05

By - Dr. Puneet Sharma Sir



Recap of previous lecture

(PARTIAL FRACTION)

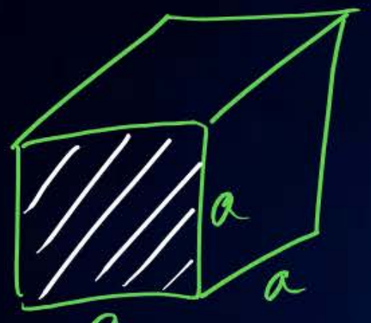


Topics *to be covered*

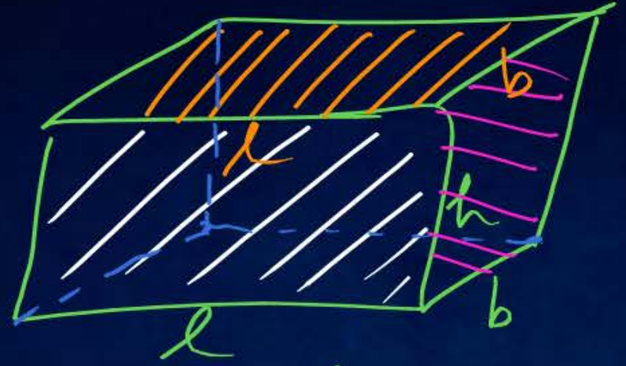
"MENSURATION"



Cube & Cuboid



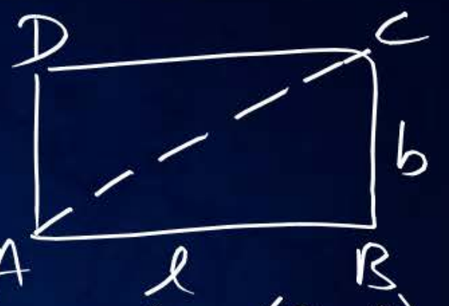
Cube

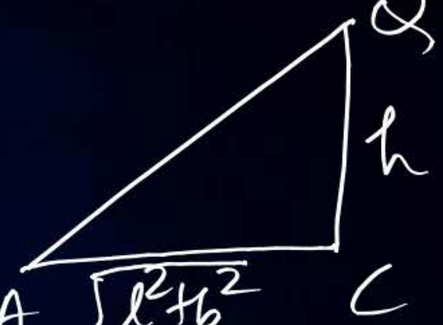


Cuboid

$AC^2 = AB^2 + BC^2$

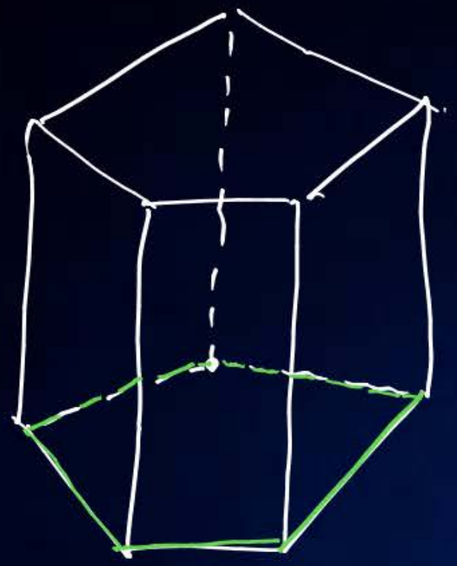
$AC = \sqrt{l^2+b^2}$





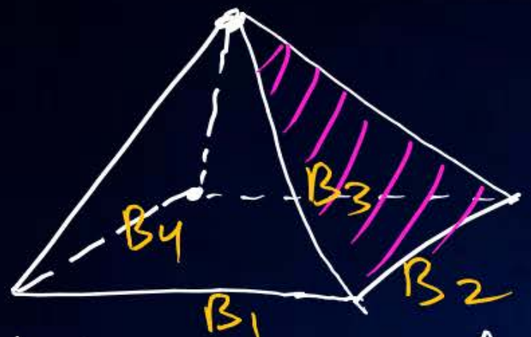
$AQ^2 = (l^2+b^2) + h^2 \Rightarrow AQ = \sqrt{l^2+b^2+h^2}$

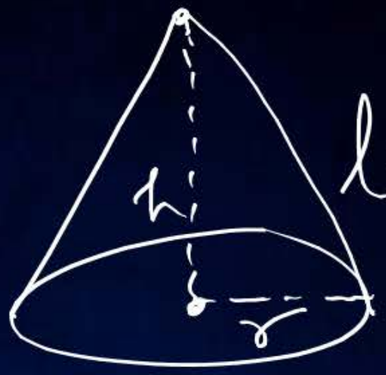
	LSA	TSA	Volume	Length of Diagonal
Cube	$4(a^2)$	$6(a^2)$	a^3	$\sqrt{a^2+a^2+a^2} = a\sqrt{3}$
Cuboid	$2(lh+bh) = 2(l+b)h$	$2(lh+bh+lb) = 2(lb+bh+hl)$	lbh	$\sqrt{l^2+b^2+h^2}$





	Right Prism	Right circular Cylinder
LSA/CSA	(Perimeter of Base) × Height	$(2\pi r)h$
TSA	2(Area of Base) + LSA	$2(\pi r^2) + 2\pi rh = 2\pi r(r+h)$
Volume	(Area of Base) × Height	$(\pi r^2)h$





l = slant Height

where $l^2 = h^2 + r^2$

	Right Pyramid	Right Circular Cone
LSA/CSA	$\frac{1}{2}$ (Perimeter of Base) × Slant Height	$\frac{1}{2}(2\pi r)l = \pi rl$
TSA	(Area of Base) + LSA	$\pi r^2 + \pi rl = \pi r(r+l)$
Volume	$\frac{1}{3}$ [Area of Base] × Height	$\frac{1}{3}(\pi r^2)h$

Proof: LSA = Area of 4 Δ's

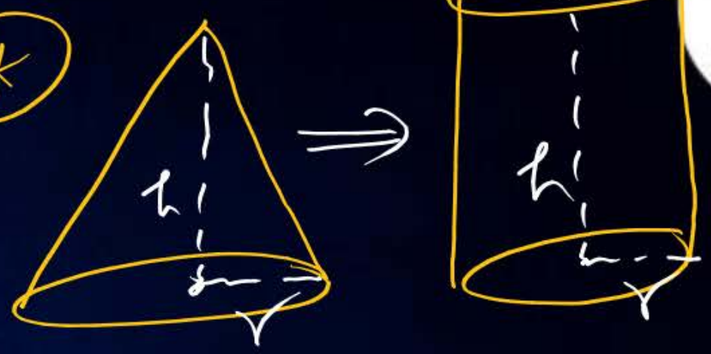
$$= \frac{1}{2} \text{ Base} \times \text{Height}$$

$$= \frac{1}{2}\left[B_1 \times SH + B_2 \times SH + B_3 \times SH + B_4 \times SH \right]$$

$$= \frac{1}{2}\left[B_1 + B_2 + B_3 + B_4 \right] \times SH$$

$$= \frac{1}{2} (\text{Perimeter of Base}) \times \text{Slant Height}$$

(*)



AT logic

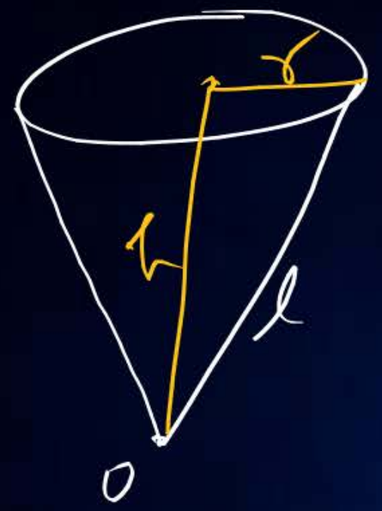
(Volume of Cone) × 3 = Vol. of cylinder

$\Rightarrow$ Vol. of Cone $= \frac{1}{3}$ (Vol of cylinder)

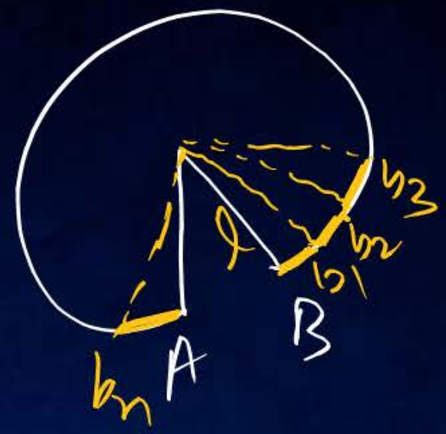
$$= \frac{1}{3}\left[\pi r^2 h \right]$$

$$= \frac{1}{3}\left[\text{Area of Base} \right] \times \text{Ht.}$$

(*)







CSA = Area of all the Δ's

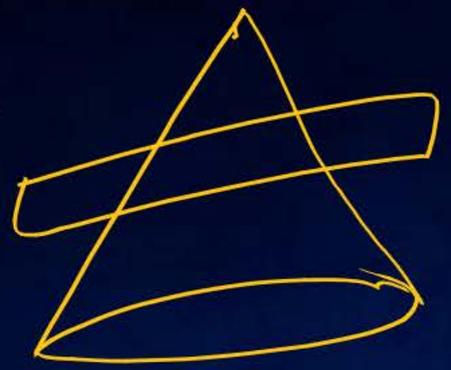
$$= \frac{1}{2}\left[b_1 \times l + b_2 \times l + b_3 \times l + \dots + b_n \times l\right]$$

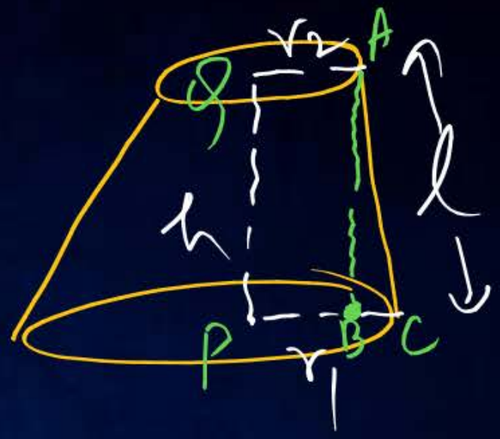
$$= \frac{1}{2}\left(b_1 + b_2 + b_3 + \dots + b_n\right) \times l$$

$$= \frac{1}{2}\left[\text{Perimeter of Base}\right] \times l$$

$$= \frac{1}{2}\left(2\pi r\right) \times l = \boxed{\pi r l}$$

Frustum of Cone :→





$AB = h$, $AC = l$

$BC = PC - QA$

$= r_1 - r_2$

In $\triangle ABC$; $AC^2 = AB^2 + BC^2$

$\boxed{l^2 = h^2 + (r_1 - r_2)^2}$

Here l = Slant Height of frustum of Cone

h = Height of " " "

CSA	TSA	Volume
$\pi r_1 l + \pi r_2 l$ $= \boxed{\pi(r_1 + r_2) l}$	$\pi r_1^2 + \pi r_2^2 + CSA$	$\frac{1}{3}\pi\left[r_1^2 + r_2^2 + r_1 r_2\right] h$

Sphere



	CSA	TSA	Volume
Sphere	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere (solid) 	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$
Spherical Shell (Concentric Spheres) 		$4\pi R^2 + 4\pi r^2 = 4\pi(R^2+r^2)$	$\frac{4}{3}\pi R^3 - \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(R^3 - r^3)$

(*)



$=$



$\Rightarrow h = 2r$

TSA of Sphere = CSA of Cylinder

$= 2\pi r h$

$= 2\pi r(2r) = 4\pi r^2$

(*)



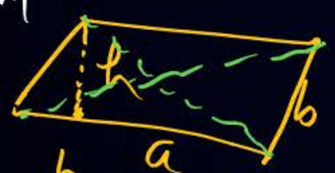


$\Rightarrow h = 2r$

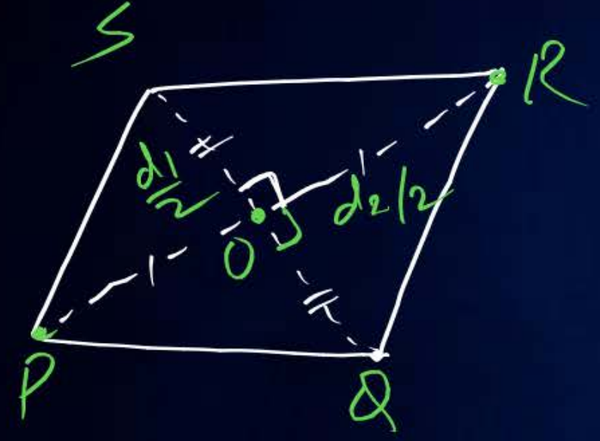
Volume of Sphere = 2 Volume of Cone

$= 2\left(\frac{1}{3}\pi r^2 h\right)$

$= \frac{2}{3}\pi r^2(2r) = \frac{4}{3}\pi r^3$

Shape	Perimeter	Area	Diagonals: EQUAL	Diagonals: BISECT	Diagonals: Intersect at 90°
Sq. 	$4a$	a^2	✓	✓	✓
Rectangle 	$2(a+b)$	ab	✓	✓	✓
Rhombus 	$4a$	$\frac{1}{2}(d_1 \times d_2)$	✗	✓	✗
∥gm 	$2(a+b)$	ah	✗	✗	✗
Trapezium 	Sum of four sides	$\frac{1}{2}(a+b)h$	✗	✗	✗

Proof of area of Rhombus :-

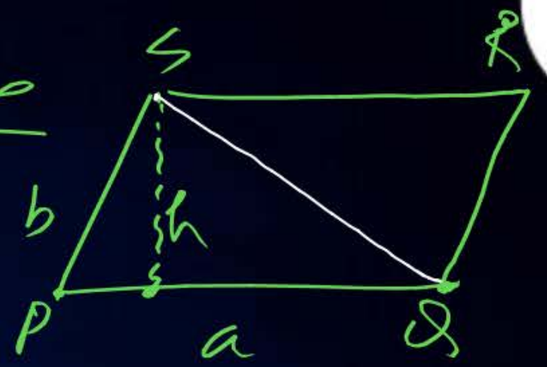


$$\text{Area} = PSR + PQR$$

$$= \frac{1}{2}\left(d_2\right)\left(\frac{d_1}{2}\right) + \frac{1}{2} d_2 \left(\frac{d_1}{2}\right)$$

$$= \frac{1}{4} d_1 d_2 + \frac{1}{4} d_1 d_2 = \frac{1}{2} d_1 d_2$$

Area of ||gm :-

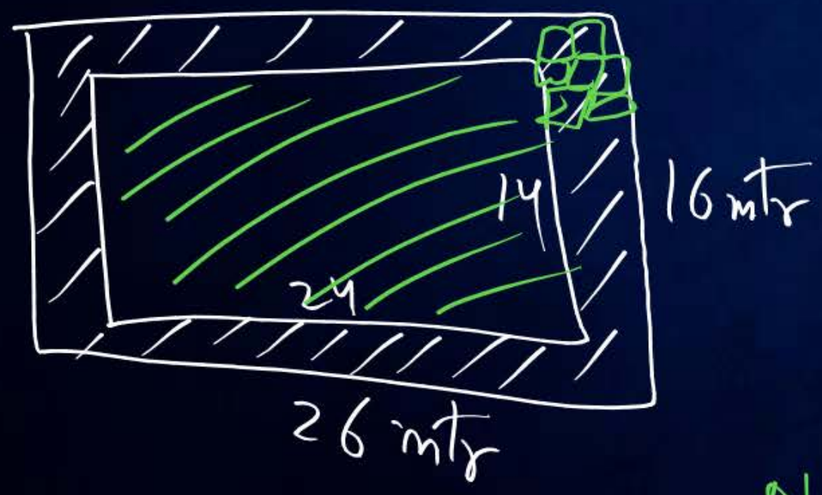


$$\text{Area} = 2(\text{Area of } PSQ)$$

$$= 2\left(\frac{1}{2} \times PQ \times h\right)$$

$$= \boxed{a \times h}$$

Qe A Garden is 24 mtr long and 14 mtr wide. There is a path 1 mtr wide outside the Garden along it's sides. If path is to be constructed with Square marble tiles of size 10 cm × 10 cm then find the number of tiles required to Cover the path? (Ans = 8000)



Area of Garden = $24 \times 14 = 336$ mtr^2

Area of (Garden + path) $= 26 \times 16 = 416$ mtr^2

Area of Path $= 416 - 336 = 80$ mtr^2

Area of one tile $= 10\text{cm} \times 10\text{cm} = 100\text{cm}^2 = \frac{100}{100 \times 100} = 0.01$ m^2

No of tiles required $= \frac{80}{0.01} = 8000$ tiles


Thank
YOU